

Section 7 - Telegram Bot Communication medium

What is Vibe coding?

Vibe coding is a new way to develop.

Its a conversational style of development.

You can build application with your natuaral langauge.



Communication Mediums for AI Agents in n8n



Context

You've built your first AI agent—now it's time to **communicate** with it. This section covers **how humans can talk to agents** using various interfaces.



Types of Communication Mediums

1. Telegram

- Easiest to set up
- Supports on-the-go interactions via mobile
- Popular in the AI agent community
- Be cautious about bot token privacy

2. WhatsApp

- Slightly harder to integrate
- Useful for business users

3. Slack

- Great for team workflows
- Takes more time to configure

4. Native n8n Chat Node

- Built-in testing tool
- Ideal for prototyping or development
- Simulates user-agent interaction via text

5. Custom Dashboards / Vibe Coding

- Advanced user interface for unique interactions
- Useful for tailored user experiences

Telegram Bot Setup Flow

1. Sign up for Telegram

- Use desktop or mobile app

2. Create Bot using BotFather

- Use command `/newbot`
- Name it and give it a unique username ending in `bot`
- Get your **Access Token** (keep this private)

3. Set up Telegram Node in n8n

- Use `On Message` trigger
- Add Telegram credentials using Access Token

4. Test the Workflow

- Send message like "Hello there"
- Use `Execute Workflow` in n8n to inspect message content

Voice Message Support (Multimodal Input)

1. Send Voice Message

- Telegram sends a `file_id` instead of text

2. Add an IF Node

- Check: Does `voice.file_id` exist?

- If **yes** → download & transcribe the voice
- If **no** → continue with normal text input

3. Get File Node

- Use Telegram's `Get a File` action
- Input `file_id` from the message

4. Transcribe Audio

- Use OpenAI's Whisper model (`Transcribe a Recording`)
- Transcribe voice into text

Send Text or Transcribed Voice to AI Agent

1. Use a Set Node

- Rename `text` or `transcribed_text` to `chatInput`
- Trick AI Agent into believing it's a chat trigger node

2. Connect to AI Agent Node

- Use OpenAI (GPT-4.1 Mini or similar)
- Prompt input source = `chatInput`
- Optionally set system instructions for custom behavior

Sending a Response Back

1. Use Telegram node again at the end
2. Map AI response back to the user via `Send Message` action

Final Setup Overview

 ~~Telegram Bot (BotFather + Access Token)~~

 ~~n8n Workflow with:~~

- Telegram On Message Trigger
- IF Node (detect voice vs text)

- Get File → Transcribe (if voice)
- Set Node to format input
- AI Agent node (OpenAI GPT)
- Telegram response message

How to Talk to Your AI Agent in n8n

Build communication layers so humans can talk to agents via messaging apps or UIs.

Communication Channels You Can Use

Medium	Setup Difficulty	Use Case	Notes
Telegram	★ Easy	Quick access, mobile-friendly	Great for personal or community use
WhatsApp	⚠ Medium	Business communication	Needs third-party API (e.g., Twilio)
Slack	⚠ Medium	Team-based workflows	Ideal for internal agents
n8n Chat Node	✅ Built-in	Testing/prototyping in dev	Good for early-stage
Custom UI	🚀 Advanced	Tailored interface (e.g., Vibe)	Best for final production

Set Up: Telegram Bot as Your First Medium

Step 1: Create a Bot via BotFather on Telegram

- Open Telegram → Search for **BotFather**
- Use `/newbot`
- Give it a name and unique username (`...bot`)
- Copy your **access token** (⚠ Keep it secret)

Step 2: Connect Telegram to n8n

- Add the **Telegram Trigger** node: `On Message`
 - Create Telegram credentials using access token
 - Test: Send "Hello" to your bot and run workflow in n8n
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Add Voice Message Support (Multimodal Input)

Want users to talk instead of text? Add voice message detection & transcription.

Step 1: Detect If the Message Is Voice

- Add **IF Node**
 - Check if `voice.file_id` exists

Step 2: Transcribe Audio

- Use `Get File` Telegram Node with `voice.file_id`
 - Download file from `file_path`
 - Use **OpenAI Whisper** to transcribe audio into text
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Connect Message to AI Agent

Step 1: Format Input

- Use **Set Node**
 - Set field `chatInput = $json["text"]` (or `transcription`)

Step 2: Pass to Agent Node

- Connect to an OpenAI (GPT-4.1 mini) or LangChain agent
- Set system instructions like:

```
vbnet
CopyEdit
```

You're a helpful AI agent that assists with XYZ

Send Response Back to Telegram

- Use `Send Message` Telegram node
 - Respond using `chat.completion.choices[0].message.content`

Final Workflow Structure

```
scss
CopyEdit
Telegram On Message
    ↓
IF (voice vs text)
    ↓ (voice)
    Get File → HTTP Request (download) → Whisper (transcribe)
        ↓
        Set Node → AI Agent (OpenAI)
            ↓
            Send Response via Telegram
```